

Analysis

Linear search goes through the list one item at a time until it finds what it's looking for. If the product is near the start, it's quick. If it's near the end, or doesn't exist, it has to check almost everything. So the more products there are, the longer it can take — the time grows along with the list size.

Binary search works differently. Since the list is sorted, it checks the middle item first and figures out if the target is bigger or smaller, then throws away half the list and repeats on the smaller half. This makes it much faster for large lists, but it only works if the data is already sorted.

For an e-commerce platform, binary search makes more sense. Products are searched very often, but the catalog itself doesn't change that frequently. So it's worth sorting the products once and then getting fast searches every time after that. Linear search would be fine only for very small lists, like items in a shopping cart, where sorting isn't really necessary.